README ▯▯▯ YouthGuard-Protocol

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# YouthGuard Protocol
**On-Chain Legal Identity & Dispute Resolution for Young Web3 Builders**
> Built for the BLI LegalTech Hackathon 2 (DoraHacks) — Team: Elias Khodadadi (Solo)
## 🧩 The Problem
Young builders in Web3 have no portable, verifiable proof of legal standing. They get scammed on bounties, excluded from opportunities for lack of credentials, and have no affordable path to dispute resolution. Existing identity solutions are either fully centralized or have no connection to real-world legal status.
## 💡 The Solution
YouthGuard anchors **legal attestations on-chain as Soulbound Tokens (ERC-5192)** and pairs them with a **decentralized dispute-resolution layer**:
1. **YouthGuardSBT.sol** — a minimal soulbound token contract. A trusted authority issues a non-transferable attestation (hash of the legal proof or ZK verification) to a user's address. Raw personal data never goes on-chain — only its cryptographic hash. Attestations expire automatically and can be revoked.
2. **LegalArbitration.sol** — an on-chain dispute layer. Anyone holding a valid YouthGuard SBT can file a dispute with an evidence hash (IPFS/ZK). Authorized arbitrators resolve or dismiss disputes, and the verdict is stored on-chain.
3. **Gateway & Agent Delegation (main.py)** — a FastAPI engine for delegating authority to AI agents *safely*: Ed25519-signed delegation tokens with a hard budget cap (USDC) and expiry, plus a full audit log of every agent action.
4. **Interactive demo (index.html)** — a bilingual (FA/EN) demo portal for judges: connect wallet, walk the 4-step protocol flow (Generate proof → Mint SBT → File case → Verdict).
## ✨ Key Features
- **ERC-5192 Soulbound attestations** — one credential per address, non-transferable, auto-expiring
- **Privacy by hashing** — only keccak256 hashes of evidence/proofs are stored on-chain, never PII
- **On-chain dispute resolution** — claimant must hold a valid SBT to file; arbitrator roster is admin-controlled
- **Agent budget control** — AI agents can act only within a signed, time-limited, capped budget envelope
- **Full audit trail** — every delegation and execution is logged and queryable
##   Tech Stack
| Layer | Technology |

|---|---|

| Smart contracts | Solidity 0.8.20, ERC-5192 (SBT) |

| Tests | JavaScript test suite (`test/YouthGuard.test.js`) |
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| Gateway / agent engine | Python, FastAPI, Ed25519 (cryptography) |

| Demo portal | HTML, Tailwind CSS (bilingual FA/EN) |

## 🚀 Quick Start
### Contracts
```bash
npm install
npx hardhat test
```

Gateway

```
pip install -r requirements.txt
uvicorn main:app --reload
```

The API docs are served at `/docs` (OpenAPI/Swagger).

🔌 API Endpoints

Method	Path	Description
GET	`/`	Health check
POST	`/api/v1/delegate`	Issue an Ed25519-signed delegation token (budget + expiry)
POST	`/api/v1/execute`	Execute a spend under a delegation token (checks expiry, agent ID, budget)
GET	`/api/v1/audit/logs`	Full audit log of agent executions

🗺️ Roadmap

- Integration of Groth16 ZK circuits for age/credential proofs (hash anchor points are already in place)
- Decentralized arbitrator staking and appeals
- NEAR-native deployment of the attestation layer

📄 License

MIT ```
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